

Classical mechanics

Small Oscillations:

Physical systems with K.E. of the form:

$$T = \frac{1}{2} T_{ij} \dot{\eta}_i \dot{\eta}_j$$

For small motion about an equilibrium point

$$q_{i(0)} : q_i = q_{i(0)} + \eta_i \quad \eta_j \ll 1$$

T_{ij} are constant $\Rightarrow T$ is a symmetric matrix.

$$\Rightarrow T_{ij} = T_{ji}$$

$$T = \tilde{T}$$

Let us expand the potential $V(q)$ around q_0 :

$$V(q) = V(q_0 + \eta) = V(q_0) + \left(\frac{\partial V}{\partial q_i} \right) \bigg|_{q_0} \eta_i$$

$$+ \frac{1}{2} \left(\frac{\partial^2 V}{\partial q_i \partial q_j} \right) \bigg|_{q_0} \eta_i \eta_j + O(\eta^3)$$

□ We are interested in motion around stable equilibrium point q_0 .

$$V = \frac{1}{2} V_{ij} \eta_i \eta_j$$

$$V_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j} \bigg|_{q_0} > 0$$

$$[V_{ij} = V_{ji}]$$

$\Rightarrow$ For stable motion around

$$L = \frac{1}{2} [T_{ij} \dot{\eta}_i \dot{\eta}_j - V_{ij} \eta_i \eta_j]$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}_k} \right) - \frac{\partial L}{\partial \eta_k} = 0 \equiv \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}_k} \right) - \frac{\partial L}{\partial \eta_k} = 0$$

$$\frac{\partial L}{\partial \dot{\eta}_k} = \frac{1}{2} T_{ij} (\delta_{ik} \dot{\eta}_j + \delta_{jk} \dot{\eta}_i)$$

$$\begin{aligned} \frac{\partial L}{\partial \dot{\eta}_k} &= \frac{1}{2} (T_{kj} \dot{\eta}_j + T_{ik} \dot{\eta}_i) \\ &= \frac{1}{2} [T_{ki} \dot{\eta}_i + T_{ki} \dot{\eta}_i] = T_{ki} \dot{\eta}_i \end{aligned}$$

$$\frac{\partial L}{\partial \eta_k} = -V_{ki} \eta_i$$

$$T_{ki} \ddot{\eta}_i + V_{ki} \eta_i = 0$$

$$\eta_i = a_i e^{-i\omega t} \rightarrow \text{oscillatory function.}$$

$\equiv$ We take either real part of this or imaginary part of it, depending on whether we need sine or cosine.

$m \ddot{\eta} + k^2 \eta = 0 \rightarrow$ we assume this function can be written as an expansion of $\sin / \cos / e^{i\omega t} \rightarrow$ Fourier series expansion.

$$\eta = \sum_{k=0}^{\infty} \eta_k e^{-i\omega t}$$

$$-\omega^2 m \eta_0 e^{-i\omega t} + \eta_0 k^2 e^{-i\omega t} = 0$$

$$\eta_0 > 0, e^{-i\omega t} \neq 0$$

$$\Rightarrow -\omega^2 m + k^2 = 0 \Rightarrow \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

n -coupled eqs $\Rightarrow$ linear.

$$-T_{ki} a_i \omega^2 e^{-i\omega t} + V_{ik} a_i e^{-i\omega t} = 0$$

$$\Rightarrow -T_{ki} a_i \omega^2 e^{-i\omega t} + V_{ki} a_i e^{-i\omega t} = 0$$

$$\Rightarrow V_{ki} a_i - \omega^2 T_{ki} a_i = 0$$

$$\Rightarrow V \cdot A - \omega^2 T A = 0$$

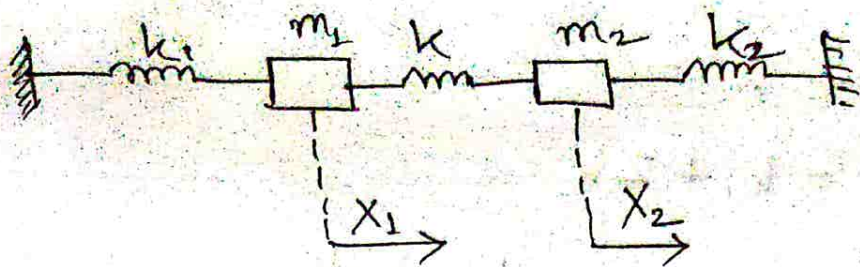
$$\Rightarrow (V - \omega^2 T) A = 0$$

Similar to eigenvalue problem:

$$(V - \lambda I) A = 0$$

$\Rightarrow$ non-trivial soln exists when

$$\det(V - \lambda I) = 0$$



$$\square T = \frac{1}{2} (m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2)$$

$$\square V = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k (x_2 - x_1)^2 + \frac{1}{2} k_2 x_2^2$$

now, $\eta_1 = x_1, \eta_2 = x_2$

$$T = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad V = \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)$$

$$V = \begin{pmatrix} k_1 + k & -k \\ -k & k_2 + k \end{pmatrix}$$

$$\begin{pmatrix} k_1 + k & -k \\ -k & k_2 + k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \omega^2 \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$m_1 = m_2 = m \quad k_1 = k_2 = k$$

$$\begin{pmatrix} k_1 + k & -k \\ -k & k_1 + k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \omega^2 m \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} k_1 + k - \lambda & -k \\ -k & k_1 + k - \lambda \end{vmatrix} = 0 \quad [P.T.O.]$$

$$\Rightarrow (k_1 + k + \lambda)^2 - k^2 = 0$$

$$\lambda = k, \quad \lambda = k_1 + 2k$$

$$\omega_1^2 = \frac{k_1}{m}, \quad \omega_2^2 = \frac{k_1 + 2k}{m}$$

$$\begin{pmatrix} k_1 + k - k_1 & -k \\ -k & k_1 + k - k_1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$= k \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

Putting another eigen value,

$$\begin{pmatrix} -k & -k \\ -k & -k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\therefore \begin{pmatrix} 1 \\ -1 \end{pmatrix} \leq \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

▣ Classical Mechanics :

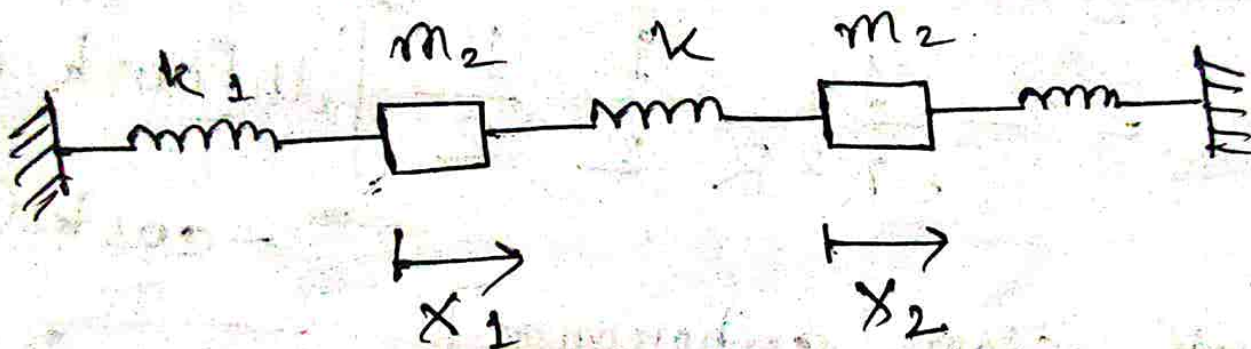
Summary: $(V_{ik} - \omega^2 T_{ik}) a_k = 0$

▣ $\omega \rightarrow$ modes of oscillation.

a_k is amplitude of oscillation,

$$\det(V_{ik} - \omega^2 T_{ik}) = 0$$

$$T_{ik} = (\) \delta_{ik}$$



$$m_1 = m_2 = m$$

$$(V_{ik} - \omega^2 m \delta_{ik}) a_k = 0$$

$$m_1 \neq m_2$$

$$T = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad V = \begin{pmatrix} k_1 + k & -k \\ -k & k_1 + k \end{pmatrix}$$

$$\Rightarrow T^{-1} = \begin{pmatrix} \frac{1}{m_1} & 0 \\ 0 & \frac{1}{m_2} \end{pmatrix}$$

$$\therefore [(V - \omega^2 m T)] A = 0$$

$$\Rightarrow \begin{pmatrix} (k_1 + k) & -k \\ -k & k_1 + k \end{pmatrix} - \omega^2 \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} A = 0$$

$$\Rightarrow \begin{bmatrix} \frac{(k_1 + k)}{m_1} & -\frac{k}{m_2} \\ -\frac{k}{m_1} & \frac{(k_1 + k)}{m_2} \end{bmatrix} - \omega^2 \mathbb{I} A = 0$$

↳ This equation here and,

$$\begin{pmatrix} k_1 + k - m_1 \omega^2 & -k \\ k & k_1 + k - m_2 \omega^2 \end{pmatrix} = 0$$

give the same result?

- Let us say, hypothetically, we have a matrix, T , which is not diagonal

$$(V - \omega^2 T) \cdot A = 0$$

$$\square S^T T S = D \Rightarrow S^T S = I = S S^T$$

$$\begin{aligned} & (S^T V - \omega^2 S^T T) S S^T A \\ &= (S^T V S - \omega^2 S^T S) S^T A \\ &= (S^T V S - \omega^2 D) S^T A \end{aligned}$$

$\triangleright$ If V is a positive symmetric matrix then V' is also a positive symmetric matrix, as it is an orthogonal transformation.

$$= [S^T V S - \omega^2 D] [A^T S]^T$$

$$= (V' - \omega^2 D) A' = 0$$

$$T = \frac{1}{2} [2\dot{x}_1^2 + 2\dot{x}_1\dot{x}_2 + 2\dot{x}_2^2]$$

$$V = \frac{1}{2} [3x_1^2 - 2x_1x_2 + 3x_2^2]$$

[Coupling occurs in both cases]

$$\therefore M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, V = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\det |V - \omega^2 M|$$

$$= \begin{vmatrix} 3-2\omega^2 & -1-\omega^2 \\ -1-\omega^2 & 3-2\omega^2 \end{vmatrix} = 0$$

$$\Rightarrow (3-2\omega^2)^2 - (-1-\omega^2)^2 = 0$$

$$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - (1 + 2\omega^2 + \omega^4) = 0$$

$$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - (1 + 2\omega^2 + \omega^4) = 0$$

$$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - 1 - 2\omega^2 - \omega^4 = 0$$

$$\Rightarrow 3\omega^4 - 14\omega^2 + 8 = 0$$

$$\Rightarrow 3\omega^4 - 12\omega^2 - 2\omega^2 + 8 = 0$$

$$\Rightarrow 3\omega^2(\omega^2 - 4) - 2(\omega^2 - 4) = 0$$

$$\Rightarrow (3\omega^2 - 2)(\omega^2 - 4) = 0$$

$$\therefore \omega = \pm 2, \pm \sqrt{\frac{2}{3}}$$

$$\therefore \omega_{1,2}^2 = 4, \frac{2}{3} \Rightarrow \omega_1^2 = 4, \omega_2^2 = \frac{2}{3}$$